XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

# CYS103 Linear Algebra

## Chapter 1

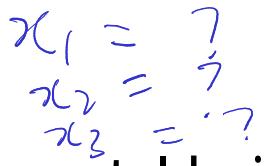
## Linear Equations and Linear Transformations

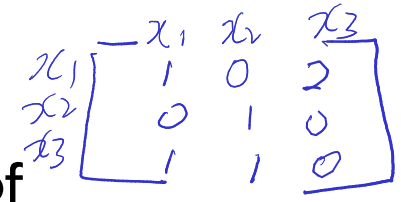
- 1.1 System of Linear Equations
- 1.2 Echelon and Reduced Echelon Forms
- 1.3 Solution Sets of Linear System
- 1.4 Linear Transformations



By Dr Yoong Kooi Kuan

# Why Do We Study Linear Algebra?




square matrix

3 x 3 matrix

We study Linear Algebra because it is one of the fundamental building blocks of

- mathematics (e.g., understanding linear systems and matrices),
- computer science (e.g., linear algebra concepts like matrix operations, eigenvalues, and transformations), and others.

To understand and solve problems (applications) by theoretical of Linear Algebra, such as:

| | |
|---|---|
| Data Science & Machine Learning | A dataset = **matrix**<br>Model training = **matrix operations** |
| Computer Graphics & Game Development | Moving a character in a game = **matrix transformation** |
| Engineering (Electrical, Mechanical) | Multiple unknowns → solved using **matrices** |

# Outline:

1.1 System of Linear Equations

1.2 Echelon and Reduced Echelon Forms

1.3 Solution Sets of Linear System

1.4 Linear Transformations

# 1.1 System of Linear Equations

**Definition**

A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ is an equation that can be expressed in the form of

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where the coefficients $a_1, a_2, \ldots, a_n$ and $b$ are real or complex numbers.

**Examples:**

Determine whether each of the following is a linear equation.

a) $3x_1 - 5x_2 = -2$

b) $4x_1 - 5x_1x_2 = 8$

c) $2x_1 + x_2 - x_3 = 2\sqrt{5}$

# 1.1 System of Linear Equations

**Definition**

A **system of linear equations** or a **linear system** is a collection of ~~one~~ **two** or more linear equations.

constant number.

**Example:**

Linear eqn. ① → $x_1 - 2x_2 + x_3 = 0$

② → $2x_2 - 8x_3 = 8$

③ → $5x_1 - 5x_3 = 10$

} system of linear equations.

① $x_1 - 2x_2 + x_3 = 0$

② $x_1 + y_1 + 3x_2 = 0$ ✗

# 1.1 System of Linear Equations

- A **solution** of the linear system, denoted as $(s_1, s_2, \dots, s_n)$, is a **list of numbers** that satisfies all equations in the system when substituted for the variables $x_1, x_2, \dots, x_n$.
- The **solution set** is the **set of all possible solutions**.

**Note:** Two linear systems are equivalent if they have the same solution set.

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**Example:**

The following is a linear system:

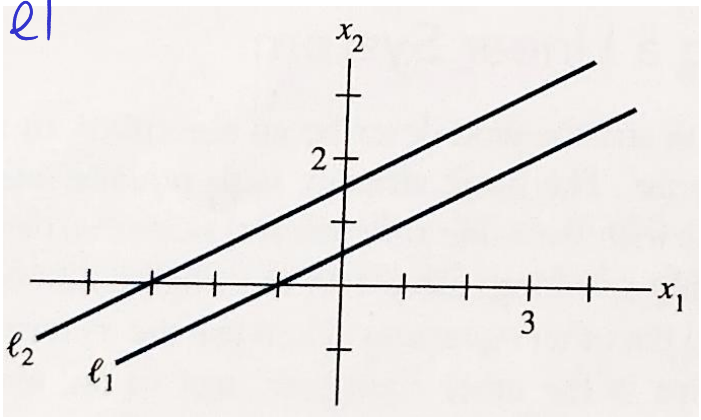
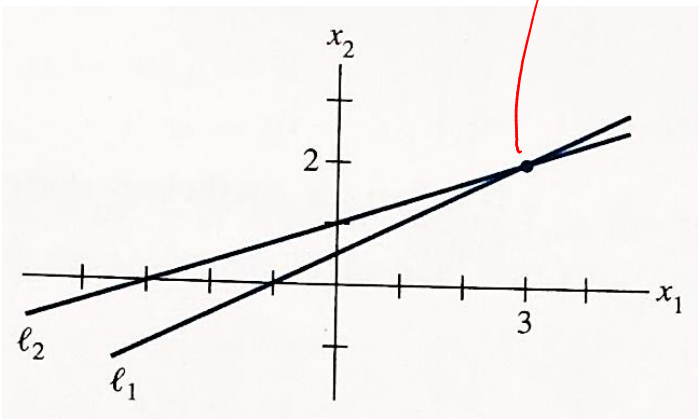
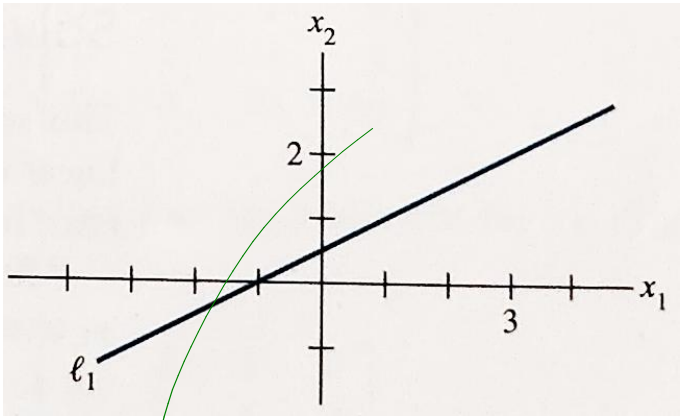
$$\begin{aligned} 2x_1 - x_2 + 1.5x_3 &= 8 \\ x_1 \quad\quad - \quad 4x_3 &= -7 \end{aligned}$$

$(5, 6.5, 3)$ is a solution of the above system for $x_1$, $x_2$ and $x_3$, respectively.

# 1.1 System of Linear Equations

an intersection point

There are three classifications of linear systems:

| No Solution | Exactly One Solution | Infinitely Many Soluitions |
|---|---|---|
|  |  |  |
| $x_1 - 2x_2 = -1$<br>$-x_1 + 2x_2 = 3$ | $x_1 - 2x_2 = -1$<br>$-x_1 + 3x_2 = 3$ | $x_1 - 2x_2 = -1$<br>$-x_1 + 2x_2 = 1$ |
| A system of linear equation is **inconsistent**. | A system of linear equation is **consistent**. | A system of linear equation is **consistent**. |

parallel

L.E ①

L.E ②

cannot find $x_1 = ?$, $x_2 = ?$

can find one solution for $x_1$ and one solution for $x_2$.

can find many sols for $x_1$ & many sols. for $x_2$.

# 1.1 System of Linear Equations

The **system of linear equations** can be written as an $m \times n$ **matrix**.

no. of rows.

no. of columns.

For example:

$$\begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$$

3×2 matrix

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}$$

linear system

matrix :

$$\begin{bmatrix} & \\ & \end{bmatrix}$$

✓ **Coefficient matrix** (or **matrix of coefficients**): $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$ 3×3 matrix

square matrix ($m = n$).

✓ **Augmented matrix**: $\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{array}\right]$

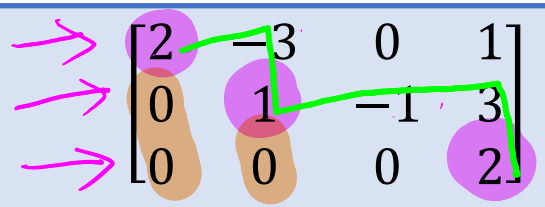
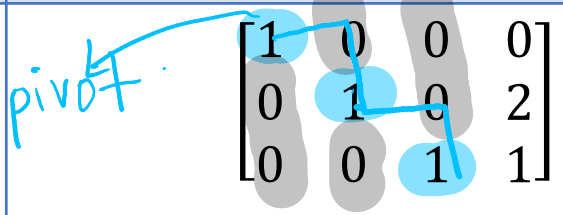
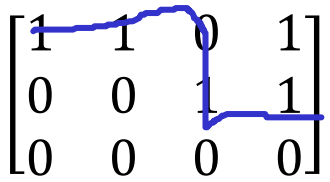
coefficients, constant no.

3×4 matrix.

# 1.2 Echelon and Reduced Echelon Forms

Reduced echelon form & Row Reduced Echelon form*

An **echelon matrix** (respectively, **reduced echelon matrix**) can be in **echelon form** (respectively, **reduced echelon form**):

| | Properties | Examples |
|---|---|---|
| **Echelon form** | • All nonzero rows are above any rows of all zero.<br>• The leading entry of each nonzero row is positioned to the right of the leading entry in the row above it.<br>• All entries in a column below a leading entry are zeros. |  <br>$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ |
| **Reduced echelon form** | • The above properties of echelon form.<br>• The leading entry in each nonzero row is 1.<br>• Each leading 1 is the only nonzero entry in its column. |  <br>  |

# 1.2 Echelon and Reduced Echelon Forms

| | Examples |
|---|---|
| **Echelon form**<br>EF. | leading entry<br>$\begin{bmatrix} 2 & -3 & 0 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$<br>leading entry.<br>$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ |
| **Reduced echelon form**<br>REF | pivot<br>$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$<br>Pivot column<br>$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ |

Note:
pivot
→ only when the leading entry is 1 in the REF.

# 1.2 Echelon and Reduced Echelon Forms

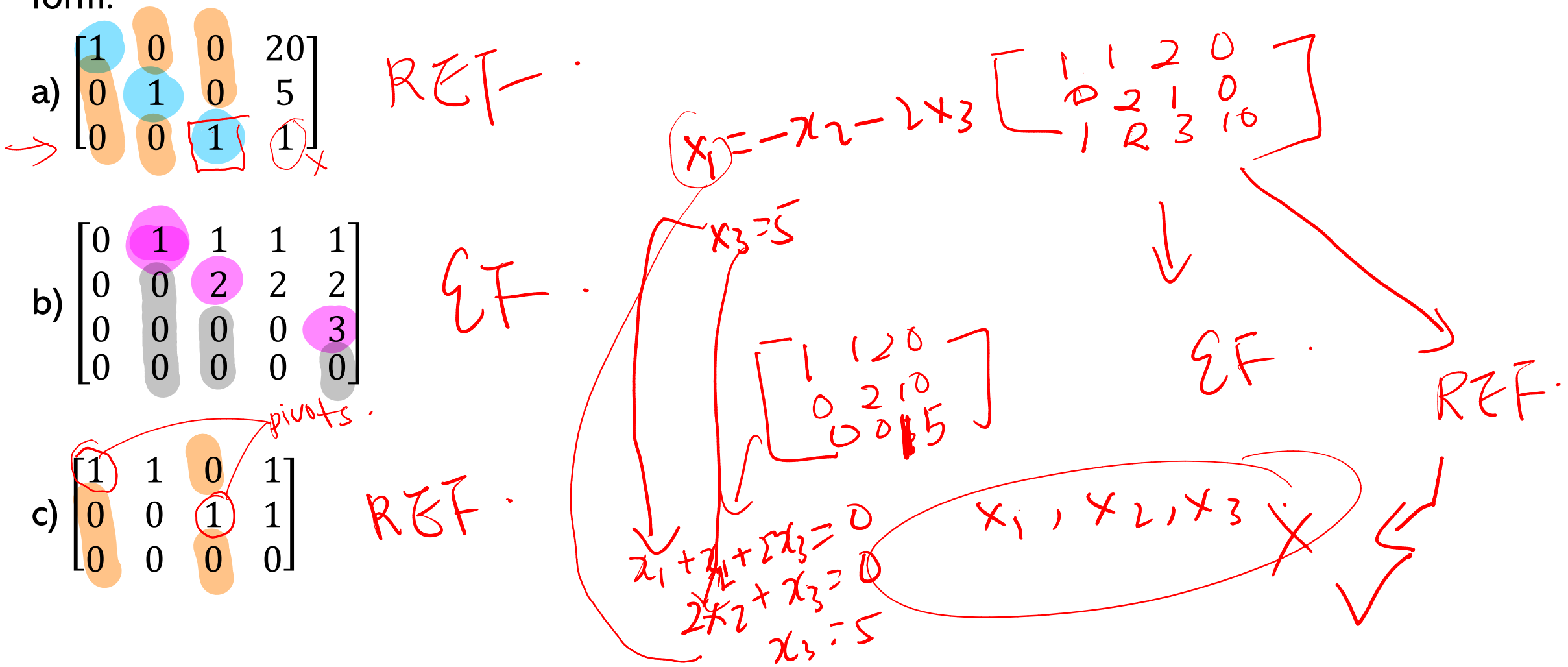
**Examples:**

Determine whether matrices are in reduced echelon form and others are only in echelon form.

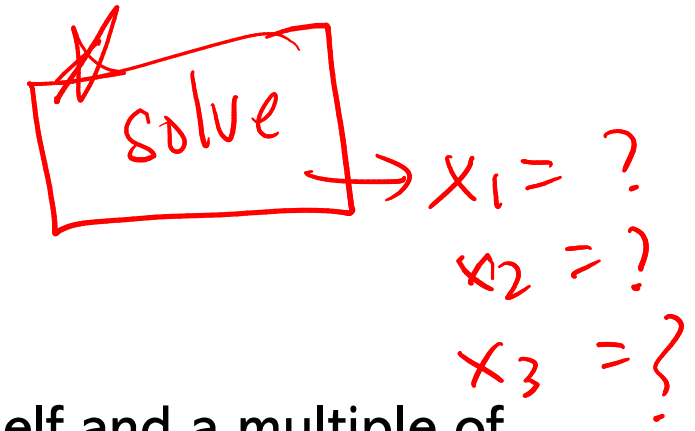
a) $\begin{bmatrix} 1 & 0 & 0 & 20 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$



# 1.1 System of Linear Equations




## Elementary Row Operations

ERO

- Solving linear systems through **Elementary Row Operations**.

1) (**Replacement**) replace one equation (or row) by the sum of itself and a multiple of another equation/row.
2) (**Interchange**) interchange two rows.
3) (**Scaling**) multiply all entries in a row by a nonzero constant.

# 1.1 System of Linear Equations

Solve.

**Example:**

Row reduced the following linear system by using **elementary row operation**.

$$x_1 - 2x_2 + x_3 = 0$$
$$2x_2 - 8x_3 = 8$$
$$5x_1 - 5x_3 = 10$$

constant number.

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix}$$

augmented matrix.

$R_3 = R_3 + (-5R_1)$ (pivot)

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & 10 & -10 & 10 \end{bmatrix}$$

$R_2 = \frac{1}{2}R_2$

$\frac{1}{2}(2)$

$\frac{1}{2}(-8)$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 10 & -10 & 10 \end{bmatrix}$$

$5 + (-5)(1) = 0$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

REF

Therefore,

$\therefore (x_1, x_2, x_3) = (1, 0, -1)$ #.

OR $x_1 = 1$, $x_2 = 0$, $x_3 = -1$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

pivot

REF!

$R_1 = R_1 + 7R_3$

$$\begin{bmatrix} 1 & 0 & -7 & 8 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$R_2 = R_2 + 4R_3$

$$\begin{bmatrix} 1 & 0 & -7 & 8 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$R_3 = \frac{1}{30} R_3$ Row 3.

$$\begin{bmatrix} 1 & 0 & -7 & 8 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 30 & -30 \end{bmatrix}$$

$R_1 = R_1 + 2R_2$

$$\begin{bmatrix} 1 & 0 & -7 & 8 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 30 & -30 \end{bmatrix}$$

$R_3 = R_3 + (-10)R_2$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 30 & -30 \end{bmatrix}$$

-10

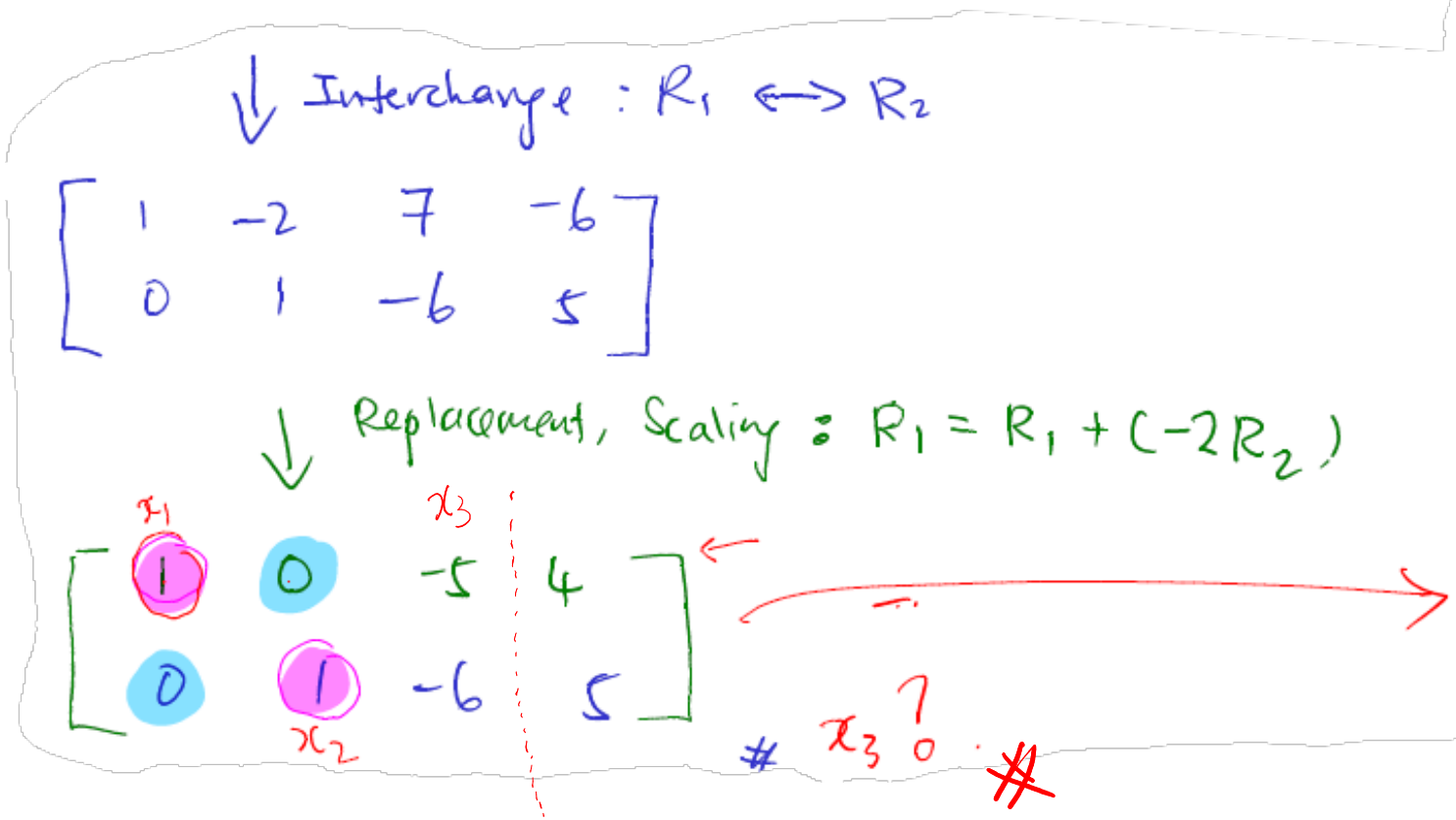
-10

# 1.1 System of Linear Equations

**Example:**

Row reduced the following augmented matrix by using elementary row operation

$$\begin{bmatrix} 0 & 1 & -6 & 5 \\ 1 & -2 & 7 & -6 \end{bmatrix}$$




augmented matrix

$\begin{bmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 10 \end{bmatrix}$

↳ Exactly one solution (consistent)

OR

infinitely many solutions!

Extra note:

example augmented matrix:

$\begin{bmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

## Extra solutions:

Thus,

$$x_1 - 5x_3 = 4$$
$$x_2 - 6x_3 = 5$$

Thus, the general solution:

$$\begin{cases} x_1 = 4 + 5x_3 \\ x_2 = 5 + 6x_3 \\ x_3 \text{ is a free variable.} \end{cases}$$

#

Note:

Basic variables: $x_1$ and $x_2$ (pivots)

free variable: $x_3$ (not pivot)

Extra note:

augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 5 \end{array}\right]$$

→ no solution!

Note:

free variable

↳ more than one exact value.

# 1.2 Echelon and Reduced Echelon Forms

**Example:**

Row reduced the following augmented matrix to reduced echelon form.

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

**Solution:**

↓ Scaling, $R_2 = \frac{1}{2}R_2$

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 1 & -4 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

↓ Replacement, Scaling : $R_1 = R_1 + (-2R_2)$

$$\begin{bmatrix} 1 & 6 & 0 & 3 & -1 & -7 \\ 0 & 0 & 1 & -4 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

$\downarrow$ Replacement, Scaling : $R_1 = R_1 + R_3$

$$\begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

$\downarrow$ Replacement, Scaling : $R_2 = R_2 + \frac{1}{2}R_3$

when the leading entries are 1 in the augmented matrix: pivots

Linear Eqn 1 : R1
" 2 : R2
" 3 : R3

$$\begin{array}{c} x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad c \\ \begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \end{array}$$

Therefore, $\begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$ is the row reduced augmented matrix in the reduced echelon form. #

To be continued.